

Totally Integrated Automation Portal

Main [OB1]

Main Properties

General

Name	Main	Number	1	Type	OB	Language	SCL
Numbering	Automatic						

Information

Title	"Main Program Sweep (Cycle)"	Author		Comment		Family	
Version	0.1	User-defined ID					

Name	Data type	Default value
▼ Input		
Initial_Call	Bool	
Remanence	Bool	
Temp		
Constant		

0001

0002 "Block_Input_Control_DB_1"();

0003

0004

0005 "Block_Indicators"();

0006

0007

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Block_Indicators [FC1]

Block_Indicators Properties

General

Name	Block_Indicators	Number	1	Type	FC	Language	SCL
Numbering	Automatic						

Information

Title		Author		Comment		Family	
Version	0.1	User-defined ID					

Name	Data type	Default value
Input		
Output		
InOut		
Temp		
Constant		
▼ Return		
Block_Indicators	Void	

```
0001 //Select Case to set range indicator.
0002 CASE "Data_block_Temps".F_TEMP OF
0003     32..49:
0004         "LL_INDC" := 1;
0005         "L_INDC"  := 0;
0006         "M_INDC"  := 0;
0007         "H_INDC"  := 0;
0008         "HH_INDC" := 0;
0009     50..99:
0010         "LL_INDC" := 0;
0011         "L_INDC"  := 1;
0012         "M_INDC"  := 0;
0013         "H_INDC"  := 0;
0014         "HH_INDC" := 0;
0015     100..149:
0016         "LL_INDC" := 0;
0017         "L_INDC"  := 0;
0018         "M_INDC"  := 1;
0019         "H_INDC"  := 0;
0020         "HH_INDC" := 0;
0021     150..199:
0022         "LL_INDC" := 0;
0023         "L_INDC"  := 0;
0024         "M_INDC"  := 0;
0025         "H_INDC"  := 1;
0026         "HH_INDC" := 0;
0027     200..212:
0028         "LL_INDC" := 0;
0029         "L_INDC"  := 0;
0030         "M_INDC"  := 0;
0031         "H_INDC"  := 0;
0032         "HH_INDC" := 1;
0033 END_CASE;
0034
```

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Block_Convert [FB1]

Block_Convert Properties

General

Name	Block_Convert	Number	1	Type	FB	Language	SCL
Numbering	Automatic						

Information

Title		Author		Comment		Family	
Version	0.1	User-defined ID					

Name	Data type	Default value	Retain
▼ Input			
C_TEMP_CONVERT	DInt	0	Non-retain
▼ Output			
F_TEMP_CONVERT	DInt	0	Non-retain
InOut			
Static			
Temp			
Constant			

0001

 //Conversion of Celsius value to Fahrenheit temperature.

0002

 #F_TEMP_CONVERT := (#C_TEMP_CONVERT* 9) / 5 + 32;

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Block_Input_Control [FB2]

Block_Input_Control Properties

General

Name	Block_Input_Control	Number	2	Type	FB	Language	SCL
Numbering	Automatic						

Information

Title		Author		Comment		Family	
Version	0.1	User-defined ID					

Name	Data type	Default value	Retain
Input			
Output			
InOut			
▼ Static			
ONE_SHOT_CONTROL	Bool	false	Non-retain
HMI_ONE_SHOT_CONTROL	Bool	false	Non-retain
INTERNAL_CONVERT	Bool	false	Non-retain
Temp			
Constant			

0001

"R_TRIG_SWITCH" (CLK:="CONVERT_PB",

0002

Q=>#ONE_SHOT_CONTROL);

0003

0004

0005

"R_TRIG_HMI" (CLK:="Data_block_HMI".HMI_CONVERT_PB,

0006

Q=>#HMI_ONE_SHOT_CONTROL);

0007

0008

0009

0010

//One shot FOR conversion.

0011

IF #HMI_ONE_SHOT_CONTROL OR #ONE_SHOT_CONTROL THEN

0012

#INTERNAL_CONVERT := 1;

0013

ELSE

0014

#INTERNAL_CONVERT := 0;

0015

END_IF;

0016

0017

0018

0019

//If no previous Errors, check for Duplicate or Over Values

0020

////and set alarms.

0021

IF #INTERNAL_CONVERT AND NOT ("ERROR_DUP_VAL") AND NOT ("ERROR_OVER_VAL")

0022

THEN

0023

"ERROR_DUP_VAL" := "Data_block_Temps".C_TEMP = "Data_block_Temps".C_TEMP_STORAGE;

0024

"ERROR_OVER_VAL" := "Data_block_Temps".C_TEMP > 100;

0025

END_IF;

0026

0027

0028

0029

//Error RESET PUSHBUTTON and clearing alarms.

0030

IF "RESET_PB" OR "Data_block_HMI".HMI_RESET_PB THEN

0031

"ERROR_DUP_VAL" := 0;

0032

"ERROR_OVER_VAL" := 0;

0033

END_IF;

0034

0035

0036

0037

0038

//IF no errors, store the Celsius value AND call the conversion routine with the Celsius temperature

0039

//TO RETURN the Fahrenheit temperature.*/

0040

IF #INTERNAL_CONVERT AND NOT ("ERROR_DUP_VAL") AND NOT ("ERROR_OVER_VAL") THEN

0041

0042

"Data_block_Temps".C_TEMP_STORAGE := "Data_block_Temps".C_TEMP;

0043

0044

"Block_Convert_DB" (C_TEMP_CONVERT:= "Data_block_Temps".C_TEMP,

0045

F_TEMP_CONVERT=> "Data_block_Temps".F_TEMP);

0046

0047

END_IF;

0048

0049

0050

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Block_Convert_DB [DB5]

Block_Convert_DB Properties

General

Name	Block_Convert_DB	Number	5	Type	DB	Language	DB
Numbering	Automatic						

Information

Title		Author		Comment		Family	
Version	0.1	User-defined ID					

Name	Data type	Start value	Retain
▼ Input			
C_TEMP_CONVERT	DInt	0	False
▼ Output			
F_TEMP_CONVERT	DInt	0	False
InOut			
Static			

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Block_Input_Control_DB [DB6]

Block_Input_Control_DB Properties

General

Name	Block_Input_Control_DB	Number	6	Type	DB	Language	DB
Numbering	Automatic						

Information

Title		Author		Comment		Family	
Version	0.1	User-defined ID					

Name	Data type	Start value	Retain
Input			
Output			
InOut			
▼ Static			
ONE_SHOT_CONTROL	Bool	false	False
HMI_ONE_SHOT_CONTROL	Bool	false	False
INTERNAL_CONVERT	Bool	false	False

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Block_Input_Control_DB_1 [DB7]

Block_Input_Control_DB_1 Properties

General

Name	Block_Input_Control_DB_1	Number	7	Type	DB	Language	DB
Numbering	Automatic						

Information

Title		Author		Comment		Family	
Version	0.1	User-defined ID					

Name	Data type	Start value	Retain
Input			
Output			
InOut			
▼ Static			
ONE_SHOT_CONTROL	Bool	false	False
HMI_ONE_SHOT_CONTROL	Bool	false	False
INTERNAL_CONVERT	Bool	false	False

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Data_block_HMI [DB1]

Data_block_HMI Properties

General

Name	Data_block_HMI	Number	1	Type	DB	Language	DB
Numbering	Automatic						

Information

Title		Author		Comment		Family	
Version	0.1	User-defined ID					

Name	Data type	Start value	Retain
▼ Static			
HMI_CONVERT_PB	Bool	false	False
HMI_RESET_PB	Bool	false	False
HMI_ENABLE_DISABLE	Bool	false	False

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Data_block_Temps [DB2]

Data_block_Temps Properties

General

Name	Data_block_Temps	Number	2	Type	DB	Language	DB
Numbering	Automatic						

Information

Title		Author		Comment		Family	
Version	0.1	User-defined ID					

Name	Data type	Start value	Retain
▼ Static			
C_TEMP	DInt	0	False
C_TEMP_STORAGE	DInt	0	False
F_TEMP	DInt	0	False